arXiv:2603.22295v1 [cs.CL] 15 Mar 2026

Whether, Not Which: Mechanistic Interpretability Reveals Dissociable Affect Reception and Emotion Categorization in LLMs

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Abstract

Large language models appear to develop internal representations of emotion—"emotion circuits," "emotion neurons," and structured emotional manifolds have been reported across multiple model families. But every study making these claims uses stimuli where emotional content is signalled by explicit emotion keywords, leaving a fundamental question unanswered: do these circuits detect genuine emotional meaning, or do they detect the word "devastated"? We present the first clinical validity test of emotion circuit claims using mechanistic interpretability methods grounded in clinical psychology. Rather than crowd-sourced text, we use clinical vignettes that evoke specific emotions through situational and behavioural cues alone, with emotion keywords systematically removed—stimuli grounded in established clinical psychology methodology. Across six models (Llama-3.2-1B, Llama-3-8B, Gemma-2-9B; base and instruct variants), we apply four convergent mechanistic interpretability methods—linear probing, causal activation patching, knockout experiments, and representational geometry—and discover two dissociable emotion processing mechanisms. *Affect reception*—the detection of emotionally significant content—operates with near-perfect accuracy (AUROC 1.000), a result consistent with early-layer saturation rather than overfitting, and replicates across all six models. *Emotion categorization*—the mapping of detected affect to specific emotion labels—is partially keyword-dependent, dropping 1–7% without keywords and improving with scale. Cross-set causal activation patching confirms that keyword-rich and keyword-free stimuli share representational space, with patched activations transferring an affective salience signal rather than emotion-category identity. These findings falsify the keyword-spotting hypothesis, establish a novel mechanistic dissociation, and introduce clinical stimulus methodology as a rigorous external standard for testing emotion processing claims in large language models. These results have direct implications for AI safety evaluation, where emotion-processing claims increasingly inform alignment and deployment decisions. All stimuli, code, and data are released for replication.

Keywords: mechanistic interpretability, emotion processing, affect reception, clinical stimuli, large language models, activation patching, representational geometry

1 Introduction

A kitchen table set for two, as usual. One plate untouched, the coffee cold. Across from her seat, his photo and a small urn…

A human reader recognizes grief instantly—not from any emotion word, but from the situation itself. A standard sentiment analyzer does not: zero emotion keywords, one incidental valence hit ("cold"—referring to coffee). Emotionally invisible.

The question this paper asks is whether a large language model sees what the human sees or what the sentiment analyzer sees.

Recent mechanistic interpretability studies paint an increasingly detailed picture of emotion processing inside LLMs. Tak et al. [1] demonstrate that emotion information consolidates in mid-layer activations and can be causally transferred via activation patching. Lee et al. [2] identify neurons that fire selectively on emotion-laden text. Reichman et al. [3] map low-dimensional emotional manifolds stable across datasets and languages. Wang et al. [4] show that prompted emotion circuits can steer model outputs. Collectively, these findings suggest that LLMs develop structured internal representations of emotion—not surface-level sentiment detection, but something closer to a functional encoding of emotional meaning.

These findings represent the first serious steps toward understanding emotional processing in LLMs. They also leave open the question this study aims to answer: are these circuits detecting emotional meaning, or detecting the keywords that co-occur with it?

Every existing study uses stimuli where emotional content is signaled by explicit emotion vocabulary. The datasets—crowd-enVENT [5], GoEmotions [6], DailyDialog [7], or synthetically generated prompts—contain phrases like "I was devastated," "she was furious," or "the news filled me with joy." When a linear probe decodes "sadness" from the residual stream of a model processing "I was devastated by the loss," it faces an ambiguity that the methodology cannot resolve: is the probe detecting the emotional meaning of the situation, or is it detecting the lexical features of the word "devastated"?

The concern is not new. Lee et al. [2] note in their discussion that identified emotion neurons may fire on cohyponyms or syntactic frames rather than emotional content per se. But nobody has run the controlled test. This paper does.

We present six LLMs with clinically designed emotional vignettes—stimuli that evoke specific emotions at controlled intensities through situational and behavioral cues, with emotion keywords systematically removed. A grief vignette describes that empty kitchen table. A rage vignette depicts scattered papers across a conference table after a falsified report surfaces. No emotion words. No sentiment phrases. No internal state descriptions. Each emotion is crossed with three distinct topic domains, controlling for the possibility that the model encodes topic rather than emotion.

We compare internal processing of these clinical vignettes to standard keyword-rich stimuli across six models (Llama-3.2-1B, Llama-3-8B, Gemma-2-9B; base and instruct variants), using four convergent methodological streams: linear probing (correlational), activation patching (causal), knockout experiments (necessity), and representational geometry (structural). This convergence within a single study is itself a methodological contribution—prior work uses one or two of these methods, never all four on the same stimulus contrast.

The results reveal a more nuanced picture than either side of the current debate anticipates. We discover two dissociable mechanisms:

1. **Affect reception**—the detection of emotional significance—is keyword-independent. Binary probes (emotional vs. neutral) achieve AUROC 1.000 on clinical vignettes across all six models, reaching ceiling in early layers. The models know that something emotionally significant is happening, from pure situational context alone.

2. **Emotion categorization**—the mapping of detected affect to specific emotion categories—is partially keyword-dependent. Eight-class probe AUROC drops 1–7% on clinical vignettes relative to keyword-rich stimuli, with larger models showing smaller drops (1.1–1.9% at 8B/9B vs. 4.6–6.7% at 1B).

We did not expect the first finding. Near-perfect binary detection without any emotion keywords—across all six models, two families, base and instruct—was surprising enough that we ran dedicated confirmation experiments to rule out artifacts. High-complexity neutral narratives matched on sensory detail and sentence structure scored 0.04 where emotional vignettes scored 0.999 (section B). Zero-shot controls confirmed the signal is intrinsic to target processing, not inherited from few-shot context. The binary probes can be trusted: what they detect is genuine emotional significance, not narrative complexity or prompt leakage.

Cross-set activation patching provides causal confirmation: patching keyword-rich activations into keyword-free forward passes transfers an *affective salience signal* that boosts categorization accuracy regardless of the source emotion's identity. This is direct evidence that affect reception and emotion categorization operate through separable causal pathways.

This paper makes four contributions. First, it provides the first clinical validity test of emotion circuit claims in LLMs, using keyword-free stimuli grounded in clinical methodology. Second, it discovers the affect reception / emotion categorization dissociation—a mechanistic distinction not previously reported. Third, it provides convergent evidence (correlational, causal, necessity, and geometric) within a single study, across six models from two families. Fourth, it releases everything—96 clinical vignettes, extraction pipeline, analysis scripts, result data—as an open-source resource for the community.